

03-May-2026

Agenda!

Feature Engineering - 6

Theory of encoding •

encoding practical

feature selection

$H(x) =$

- $B \rightarrow$
- $A \rightarrow$
- $C \rightarrow$
- $D \rightarrow$
- $B \rightarrow X$
- $A \rightarrow X$
- $C \rightarrow X$
- $D \rightarrow X$

.unique()

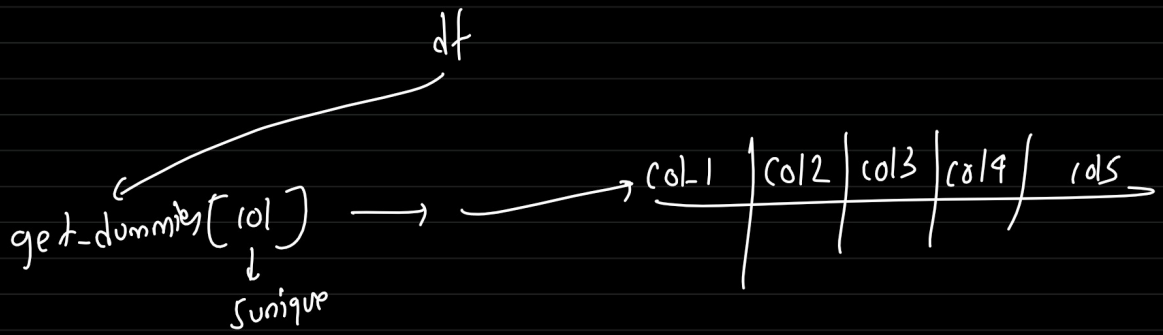
$[B, A, C, D]$

	Month()	Month-J	Month-F	Month-M	Month-Apr	Month-May	Month-Jun	Month-Jul	Month-Aug	Month-Sep	Month-Oct	Month-Nov	Month-Dec
0	J	1	0	0	0	0	0	0	0	0	0	0	0
1	F	0	1	0	0	0	0	0	0	0	0	0	0
2	M	0	0	1	0	0	0	0	0	0	0	0	0
3	Apr	0	0	0	1	0	0	0	0	0	0	0	0
4	May	0	0	0	0	1	0	0	0	0	0	0	0
5	Jun	0	0	0	0	0	1	0	0	0	0	0	0
6	Jul	0	0	0	0	0	0	1	0	0	0	0	0
7	Aug	0	0	0	0	0	0	0	1	0	0	0	0
8	Sep	0	0	0	0	0	0	0	0	1	0	0	0
9	Oct	0	0	0	0	0	0	0	0	0	1	0	0
10	Nov	0	0	0	0	0	0	0	0	0	0	1	0
11	Dec	0	0	0	0	0	0	0	0	0	0	0	1

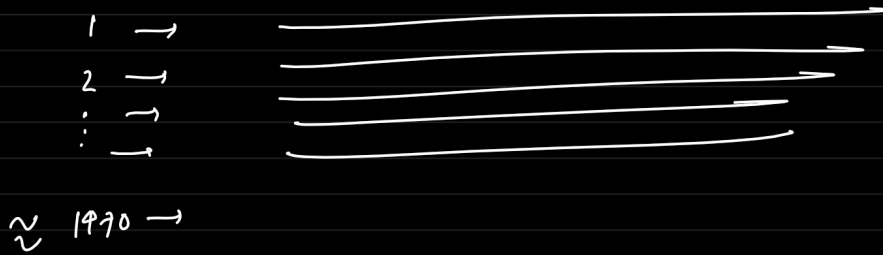
$\rightarrow$ OHE

(100000)

$7 > 0$



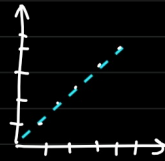
df + col_1 + col_2 + col_3 + col_4 + col_5 → deep d.



feature selection

- Backward elimination
 - Forward elimination
 - RFE → Recursive feature elimination
- ML

A	B
1	2
2	4
3	6
4	8
5	10



$$y = mx$$

- ① mathematically
- ② Machine learning

$$x_1 \rightarrow y = mx$$

$$x_1, x_2 \rightarrow y = m_1 x_1 + m_2 x_2$$

$$x_1, x_2, x_3 \rightarrow y = m_1 x_1 + m_2 x_2 + m_3 x_3$$

↓
linear equation

$$\begin{matrix} x_1 & m_1 \\ x_2 & m_2 \\ x_3 & m_3 \end{matrix}$$

↓
 m_i decides how much influence of x_i to predict y

↓

$$m_1 x_1 + m_2 x_2 + m_3 x_3$$

$$\rightarrow m_1 = 0$$

that x_1 has no influence

↙
 m_1, m_2, m_3 in linear equations define the importance of the feature

our example:-

let suppose we have 4 feature, 1 target (sale price)

ML $\rightarrow$ y

↓
 $[m_i] \rightarrow \text{len}(40)$

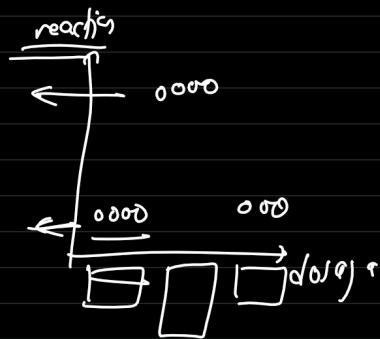
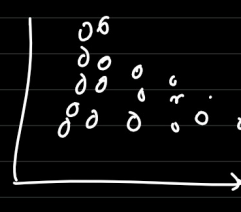
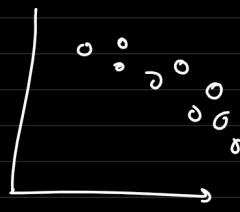
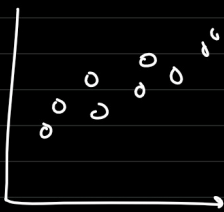
Data $\rightarrow$ linear
 $\rightarrow$ non-linear $\rightarrow$ ML $\rightarrow$ linear Reg

↳ ML $\rightarrow$ Random Forest Reg (Tree)
complexity

↓
Most powerful models

↓

xgboost $\rightarrow$
random forest



Backward Elimination: (statistical)

- ① ML $\rightarrow$ linear regression
 $\rightarrow$ Tree algorithms
- ② x & y defined
- ③ loop: [10] (old) $\rightarrow$ p-value
 $\rightarrow [x] \rightarrow ML \rightarrow y$
 $\rightarrow$ p-value of each & every column
 $\rightarrow \max(pvalue) \rightarrow$ id column
 $\rightarrow \max(pvalue) > \text{threshold:}$
 $\rightarrow \text{remove []} \rightarrow$ column from $x[]$
 $\rightarrow$ only column

H_0 : feature is not useful
 $\therefore m \rightarrow 0$

H_1 : feature is useful

hypothesis testing, α

$p < \alpha \rightarrow$ useful

else $\rightarrow p > \alpha$: not useful

one-tail

while $\rightarrow$ loop

break $\rightarrow$ no feature to drop

start all features, fit a regression & remove least significant.

Forward selection : (statistical)

(1) p-value
loop:

$x[0]$ + add one column from $X \rightarrow ML \rightarrow y$

if $p < \text{threshold}$:
add feature to $x[]$

$\rightarrow x[n] \rightarrow \text{significant} \rightarrow ML$

(2) R^2

RFE: p-value / $R^2 \rightarrow X$

$\rightarrow$ model based selection

$\downarrow$

Ranking $\rightarrow$ like Backward Elimination